

ChargePe — Smart EV Charging Platform

A **modern, frontend-only EV charging ecosystem** that combines a **real-time user app** with a **live admin analytics dashboard**. Built to simulate how a production-grade EV network behaves — bookings, charging, analytics, invoices, maps, and sync — all without a backend.

What is ChargePe?

ChargePe is an **interactive EV charging experience** designed for demos, portfolios, and academic projects. It visualizes how users find stations, reserve slots, charge vehicles, and how admins monitor revenue, sessions, traffic, and performance — all in real time.

 *Think of it as a complete EV startup frontend — map, bookings, analytics, and dashboards included.*

Project Architecture

```
User App (index.html)
├─ Live Map (Leaflet)
├─ Station Discovery
├─ Booking & Charging Simulator
├─ AR Overlay
├─ Invoice Generator
└─ Local Storage (Bookings)
    └─ Real-Time Sync
Admin Dashboard (dashboard.html)
├─ Live KPIs
├─ Charts & Analytics
├─ Traffic Sources
├─ Activity Logs
└─ CSV / PDF Export
```

 **Real-time sync** is handled entirely via `localStorage` + cross-tab events.

Key Features (Visualized)

User App — EV Charging Experience

-  **Live Interactive Map** (Leaflet)

- Auto-generated EV stations (50–350 kW)
- Location search + nearby station detection
- Station status: Available / Charging / Reserved
- 🚢 **Slot Reservation & Booking Flow**
- **Charging Simulator** (battery %, kWh, power rate)
- **PDF Invoice Generation** (jsPDF)
- Export / Import bookings (JSON)
- Smart ETA & distance calculation
- 🏗️ Dark / Light theme toggle
- Fully responsive (mobile → 4K)
- 🚢 **AR Overlay mode** (visual demo layer)
- 🚗 Offline-first behavior (PWA-ready)

🏠 *Feels like a real charging session — but runs fully in the browser.*

🐘 Admin Dashboard — Real-Time Analytics

- 🏠 **Live KPI Cards**
 - Sessions
 - Revenue (₹)
 - Customers
- 🏠 Interactive Charts (Chart.js)
 - Reports (Today / Month / Year)
 - EV Charging Trends
 - Traffic Sources
- 🚦 Date Filters
 - Today
 - Week
 - Month
 - Quarterly
 - Yearly
- 🚢 **Recent Bookings Table**
 - Live booking popup notifications
 - Activity logs
 - Export Reports
 - CSV (PapaParse)
 - PDF (jsPDF + AutoTable)
 - 🏗️ Dark / Light mode

 Open dashboard + user app in two tabs and watch data update instantly.

Real-Time Sync Engine

Files Involved

- `chargepe-sync.js`
- `chargepe-realtime-sync.js`

What It Does

- Hooks into `addBooking()` automatically
-  Syncs bookings from **User App** → **Admin Dashboard**
-  Auto-generates demo bookings every 30 seconds
- Normalizes data (customers, revenue, duration)
- Prevents storage overflow (20k-25k cap)
-  Works across tabs — **no refresh needed**

No backend. No sockets. Still feels live.

Tech Stack

Layer	Technology
UI	HTML5, CSS3, Glassmorphism
Maps	Leaflet.js, Routing Machine
Charts	Chart.js
Storage	localStorage
Sync	Storage Events
Export	jsPDF, AutoTable, PapaParse
Animations	Lottie
Icons	Font Awesome

Project Structure

```
ChargePe
├── index.html          # User EV App
└── dashboard.html     # Admin Dashboard
```

```
├ style.css # High-end responsive system
├ chargepe-sync.js # Booking sync (basic)
└ chargepe-realtime-sync.js # Advanced real-time sync
```

How to Run

1. Clone or download the project
2. Open `index.html` in browser
3. Open `dashboard.html` in another tab
4. Make a booking or wait 30 seconds
5. Watch analytics update instantly 🏠

No server No database No setup

Use Cases

- 🏢 Portfolio / Resume project
 - Academic demonstrations
 - 🏠 UI / UX experimentation
 - EV startup concept demo
 - 🏠 Analytics dashboard showcase
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Future Enhancements (Ideas)

- 🚚 Auth & role-based access
 - Backend (Firebase / Supabase)
 - 🌐 WebSocket real-time sync
 - AI-based demand prediction
 - 🏙️ City-wise heatmaps
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Author

Built with and by **Vinay**

ChargePe shows how far frontend engineering alone can go.

★ If you like this project, give it a star and plug in! 🐘